



TymeX Accelerates Clean Coding by 40% by Implementing Generative AI on AWS

TymeX improved developer productivity with Amazon CodeWhisperer and Amazon Q, automating tasks across the software development lifecycle while using real-time AI assistants to troubleshoot errors.

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40%

less time spent on code writing and te



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



90%

increase in testing efficiency, from 5 hours to 50 minutes



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



77%

AI mindset

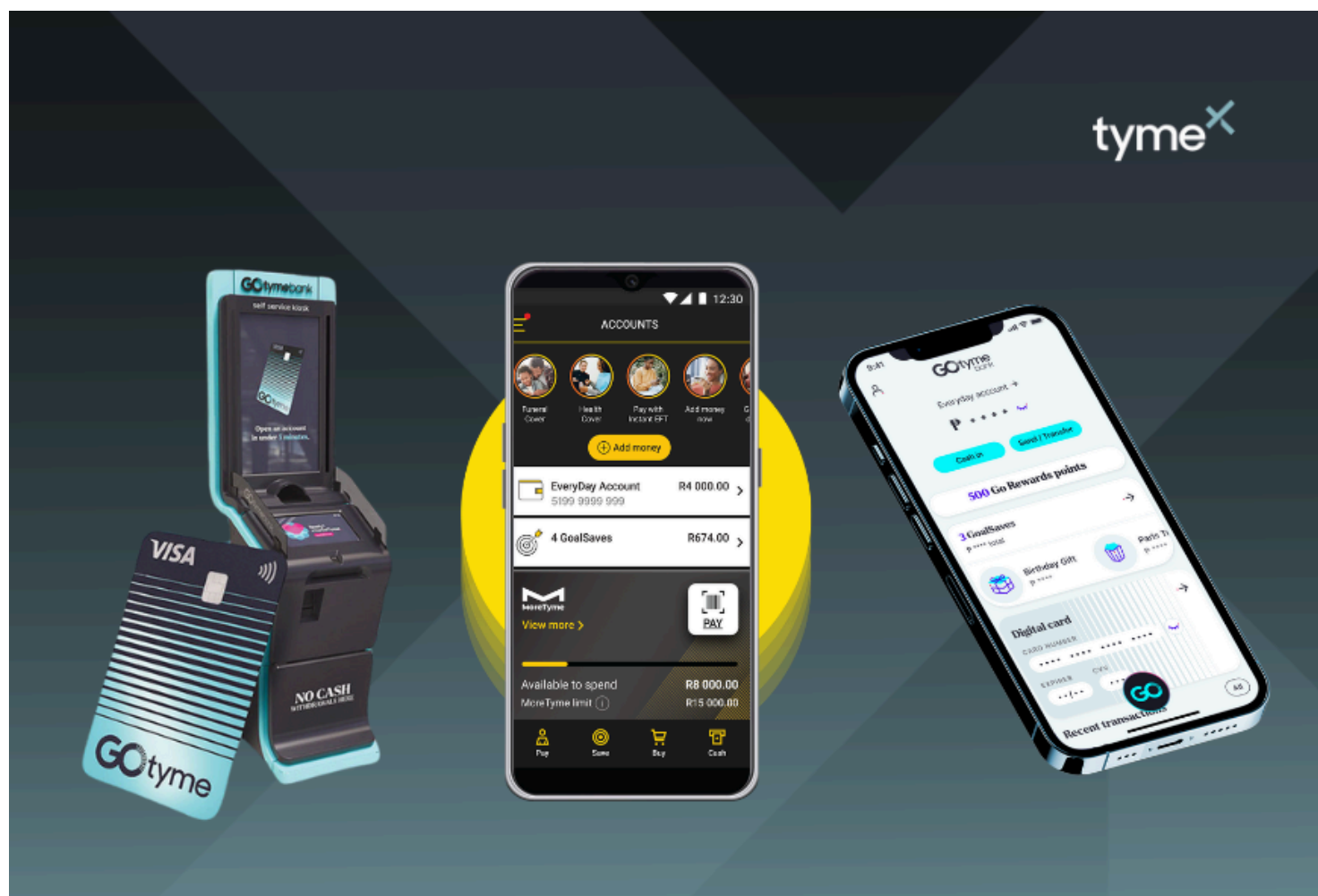
to summarize 100-page documents



Overview

[TymeX](#) builds scalable, replicable digital banks for underserved and underbanked populations. To ensure the security of its platform, the company chose to implement and develop generative AI models on AWS.

TymeX is using [Amazon CodeWhisperer](#) to automate aspects of the software development lifecycle, [Amazon Q](#) as a real-time AI assistant, and [Amazon Bedrock](#) to develop custom generative AI applications. With the enhanced automation, developers are 40 percent more productive, can assimilate complex documentation faster, and have more time to devote to innovation and skill development.



Opportunity | Building an AI-First Engineering Culture

TymeX Group is reimagining digital banking in emerging markets. Its TymeX division offers a scalable and highly replicable model for designing, building, and commercializing digital banks for underserved populations. Deployed first in South Africa and more recently in the Philippines, TymeX currently serves 12 million customers.

TymeX has long valued an agile development culture, with an emphasis on efficient, quick iteration. The company also recognizes the importance of employee satisfaction in delivering the best product possible and cultivating innovation. In 2023, the business began experimenting with generative artificial intelligence (AI) to streamline its development workloads.

Tri Tran, solution architect at TymeX, says, “We believe in working closely with our internal teams to understand their challenges, identify opportunities for AI integration, and cocreate solutions that address employees’ needs effectively. At TymeX we’re building an AI-first engineering culture. AI isn’t just an extra tool or side project; it’s at the heart of how our organization operates.”

Solution | Automating Recurring Tasks, Developing AI Chatbots

Given the industry context of banking, ensuring security and compliance for AI projects was essential. After exploring various platforms, TymeX approached Amazon Web Services (AWS) for support in developing generative AI applications. Says Tri, “AWS offers a wide range of security features and services that meet the most stringent industry standards, and its platform is trusted by countless financial institutions across the world.”

TymeX had a clear implementation strategy: it would first purchase and customize generative AI solutions on AWS, facilitating quick deployment of new technology in its operations. Phase two involved building custom generative AI applications in-house. Collaborating with experts from [AWS Enterprise Support](#), TymeX honed in on an initial use case: automating recurring aspects of its software development lifecycle (SDLC). This would help developers focus on more critical business logic—while simultaneously enhancing code quality.

AWS conducted on-site training sessions in cloud networking and introduced Amazon CodeWhisperer, an AI-powered productivity tool. TymeX applied Amazon CodeWhisperer in the SDLC to help over 200 developers write unit tests. The service improves productivity by generating code recommendations from natural-language questions, saving time developers would spend searching for code snippets. It also completes boilerplate code automatically, suggests entire functions or blocks from comments, and proactively detects security issues. TymeX also implemented Amazon Q, an interactive, generative AI-powered assistant that supports developers by answering questions about coding tasks, AWS services best practices, and internal documentation. “Amazon Q helps us with troubleshooting errors as we deploy various services on AWS, explaining why the error might have occurred and providing suggestions on how to fix it,” explains Tri.

Tri continues, “Amazon CodeWhisperer and Amazon Q form a powerful duo. Developers have help with every part of their job, from writing code to deploying solutions. These tools boost productivity, reduce errors, and



learning curves, and ensure developers can focus on their most creative work.”

TymeX chose Amazon Bedrock to develop its own AI solutions, utilizing its diverse range of AI models. Engineers built an internal AI chatbot called Tymee to answer questions about company directories and more. The chatbot soon made its way onto Slack, TymeX’s internal communication channel, to promote AI adoption throughout the company. Employees can ‘like’ or ‘dislike’ answers received from Tymee, which helps the AI bot learn and improve over time. The chatbot was recently applied to a new use case in human resources, helping non-technical employees screen curricula vitae for engineering and development roles.

Outcome | Improving Efficiency with Higher Customer and Employee Satisfaction

Since implementing these AI-powered initiatives, TymeX has seen significant improvements in efficiency, collaboration, and employee well-being. Developers are completing their work 40 percent faster, which in turn helps them allocate more time for innovation and feature delivery. Additionally, they have more time to learn new programming languages and advance their careers.

“Our developers now have more opportunities to update their skills and knowledge on technology trends whether as part of our strategic business direction or their personal development plans,” says Tri. In addition, the Tymee chatbot has received an overall positive feedback score of 77 percent from internal users.

Another AI application is TymeX’s chat-to-document feature, where an AI chatbot helps users understand complex information by summarizing lengthy documents quickly—processing 100 pages in under 10 seconds. This feature supports compliant software design, essential for managing the extensive technical and compliance documentation required to launch financial institutions internationally. TymeX has also integrated AI-driven threat detection and prevention mechanisms to fortify its security posture.

Engineers and developers at TymeX are making more data-driven decisions, which has reduced the volume of coding errors and accelerated the deployment of new software solutions.

They can now perform unit tests more efficiently, achieving 89 percent code coverage in just 30 minutes—a significant reduction from the previous 5 hours required. This represents a 90 percent increase in testing efficiency, greatly enhancing the speed and quality of software development. TymeX built its first digital bank in three years, its second in 18 months—and it expects to launch a third location by the end of 2024, in just nine months.

“Generative AI has transformed how we work and deliver value to our customers. With the best practices and knowledge we’ve acquired, we’re fueled to deliver tailored AI solutions to meet business needs,” concludes Tri.

Learn More

To learn more, visit [generative AI on AWS](#)

About TymeX

TymeX is Tyme Group’s technology and product development hub with a mission to become a global leader. The company creates scalable, replicable digital banking models for underserved and underbanked populations.



AWS Services Used

Amazon CodeWhisperer

Your AI-powered productivity tool for the IDE and command line.

[Learn more »](#)

Amazon Q

The most capable generative AI-powered assistant for accelerating software development and leveraging companies' internal data.

[Learn more »](#)

Amazon Bedrock

The easiest way to build and scale generative AI applications with foundation models.

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AWS Enterprise Support

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